



Digital Egypt Pioneers Initiative Fortinet Project Implementing VPN Solutions with FortiGate

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Presented To:

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Agenda

- **Introduction**
- **Week 1 Deliverables**
- **Week 2 Deliverables**
- **Week 3 Deliverables**
- **Conclusion**
- **Q&A**

Overview of VPNs and Their Importance

A Virtual Private Network (VPN) is a secure connection that allows users to access a private network over the internet. VPNs encrypt data traffic, ensuring confidentiality, integrity, and anonymity. They are crucial for businesses and individuals as they:

- Protect sensitive data from cyber threats.
- Enable secure remote access for employees.
- Ensure privacy when using public networks.
- Facilitate safe communication between branch offices or across geographies.

- A leading global provider of cybersecurity and networking solutions
- Established in 2000, trusted by enterprises, governments, and service providers
- Creator of FortiGate, one of the most widely deployed next-generation firewalls

Solution Areas

- Network security (NGFW, IPS, Web Filtering)
- Secure remote access & VPN technologies
- SD-WAN and branch connectivity
- Cloud and data center protection



Why FortiGate for VPN Solutions

FortiGate, developed by Fortinet, is a leading network security platform that offers robust VPN solutions. It is widely chosen because it:

- Provides high-performance, hardware-accelerated VPN connections.
- Supports both SSL and IPsec VPNs for flexible deployment.
- Integrates advanced security features like firewall, intrusion prevention, and malware protection.
- Offers easy management through a centralized interface, making VPN deployment and monitoring efficient.



Project Stages

1- SSL VPN Configuration

- Remote access for users
- Authentication, portal settings, and access policies

2- IPsec Site-to-Site VPN

- Private communication between two networks
- Routing and traffic selectors to enable encrypted connectivity

3- SD-WAN Integration

- Link performance monitoring
- Intelligent VPN traffic steering for reliability and continuity

Week 1 – VPN Concepts & SSL VPN Configuration on FortiGate

What is a VPN?

A Virtual Private Network (VPN) is a technology that creates a secure, encrypted tunnel between a user and a network over the internet.

Why do we use a VPN?

- Protects data from interception
- Allows safe remote access to internal resources
- Connects networks securely over public channels
- Ensures privacy and confidentiality

Types of VPNs

Site-to-Site VPN (Network ↔ Network)

- Connects entire networks (e.g., HQ ↔ branch) .
- Used for permanent connections .
- Ideal for businesses with multiple locations .
- Runs mainly on IPsec .
- Always-on and transparent to users .

Remote Access VPN (User ↔ Network)

- Allows individual users to connect remotely .
- Ideal for employees working from home/outside .

Two main types :

SSL VPN

IPsec Remote Access

Requires user authentication .

SSL VPN vs IPsec VPN

• **SSL VPN**

Works over HTTPS (TCP 443)

Uses SSL/TLS encryption

Easy to access (browser/FortiClient)

Works smoothly behind NAT & firewalls

Best for remote users / work from home

• **IPsec VPN**

Uses IPsec protocols .

Requires a VPN client .

More secure for managed corporate devices .

Very stable for site-to-site tunnels .

Preferred for permanent office connections .

Configuration (Week 1 – SSL VPN)

Step 1 – Interface Configuration:

- Port1 (WAN) was configured to receive a DHCP IP address for external connectivity.**
- Port2 (LAN) was assigned the static subnet 192.168.100.1/24 to serve internal hosts.**

Interfaces were verified to ensure proper WAN and LAN separation.

FortiGate-VM64

Dashboard

Network

Interfaces

DNS

Packet Capture

SD-WAN

Static Routes

Policy Routes

RIP

OSPF

BGP

Routing Objects

Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

FortiGate VM64

1 3 5 7 9 11 13 15 17 19 21 23

2 4 6 8 10 12 14 16 18 20 22 24

Create New Edit Delete Integrate Interface Search

Group By Type

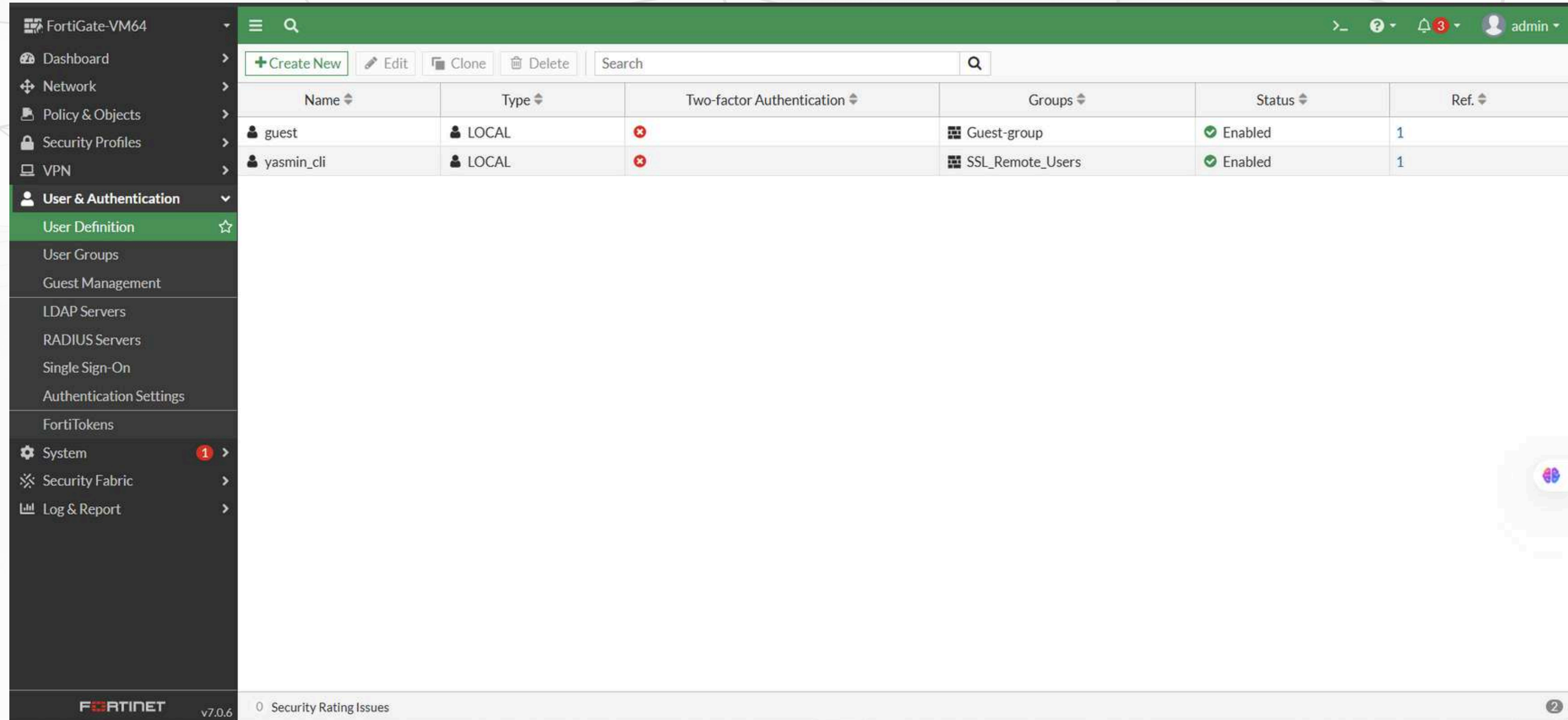
Name	Type	Members	IP/Netmask	Administrative Access	DHCP Clients	DHCP Ranges	Ref.
802.3ad Aggregate 1							
fortilink	802.3ad Aggregate		Dedicated to FortiSwitch	PING Security Fabric Connection		10.255.1.2-10.255.1.254	2
Physical Interface 10							
port1	Physical Interface		192.168.198.132/255.255.255.0	PING HTTPS SSH HTTP FMG-Access			1
port2	Physical Interface		192.168.100.1/255.255.255.0	PING HTTPS HTTP	1	192.168.100.50-192.168.100.200	4
port3	Physical Interface		0.0.0.0/0.0.0.0				0
port4	Physical Interface		0.0.0.0/0.0.0.0				0
port5	Physical Interface		0.0.0.0/0.0.0.0				0
port6	Physical Interface		0.0.0.0/0.0.0.0				0
port7	Physical Interface		0.0.0.0/0.0.0.0				0
port8	Physical Interface		0.0.0.0/0.0.0.0				0

0 Security Rating Issues

0% 12 Updated: 09:12:36

Step 2 – Local User Creation:

- A local user (yasmin_cli) was created for SSL-VPN authentication.
- The user was assigned a secure password and enabled for login.



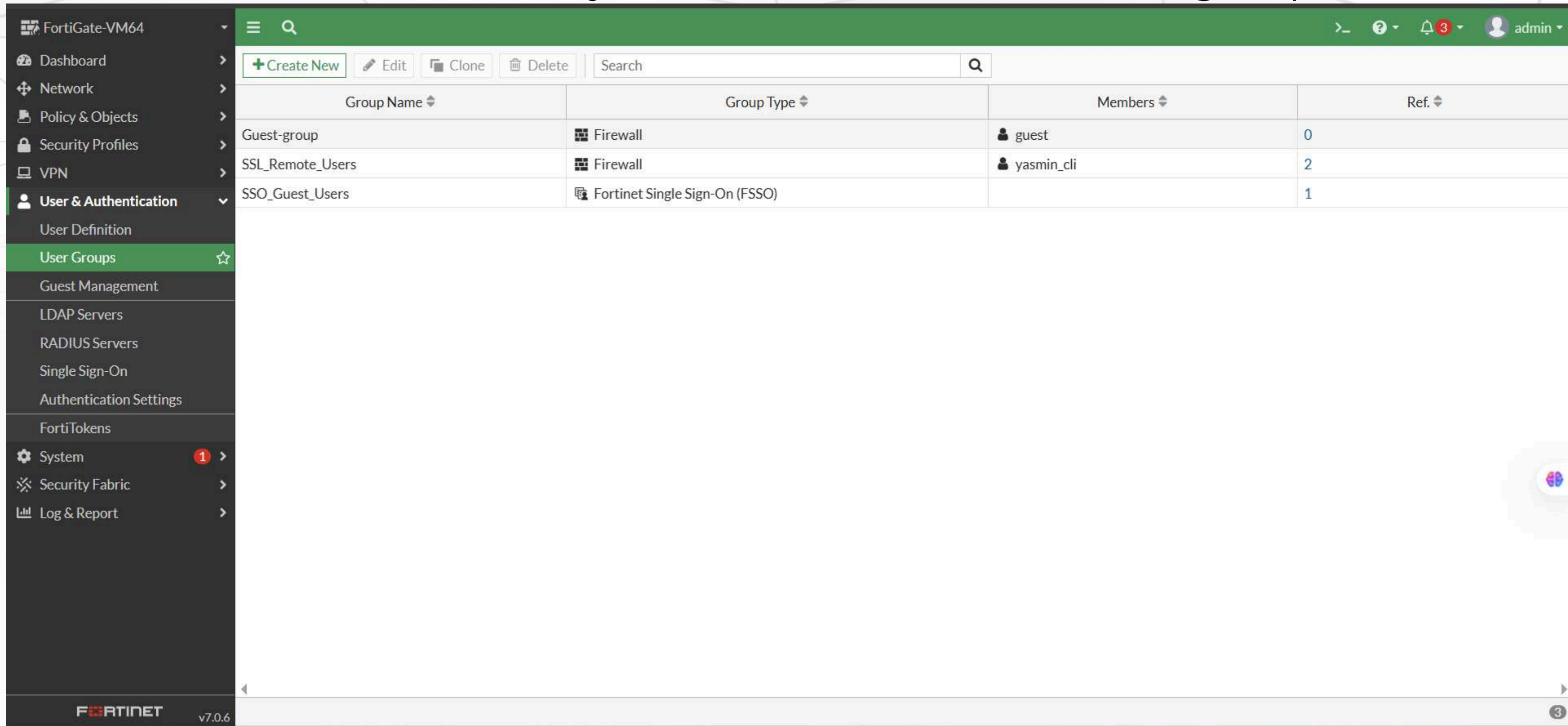
The screenshot shows the FortiGate-VM64 User & Authentication interface. The left sidebar contains the navigation menu with 'User & Authentication' expanded. The main content area displays a table of user definitions.

Name	Type	Two-factor Authentication	Groups	Status	Ref.
guest	LOCAL	✗	Guest-group	✓ Enabled	1
yasmin_cli	LOCAL	✗	SSL_Remote_Users	✓ Enabled	1

The interface also includes a top navigation bar with 'Create New', 'Edit', 'Clone', and 'Delete' buttons, and a search bar. The bottom status bar shows 'FORTINET v7.0.6' and '0 Security Rating Issues'.

Step 3 – User Group Configuration:

- A dedicated group named SSL_Remote_Users was created.
- The local user yasmin_cli was added to this group.

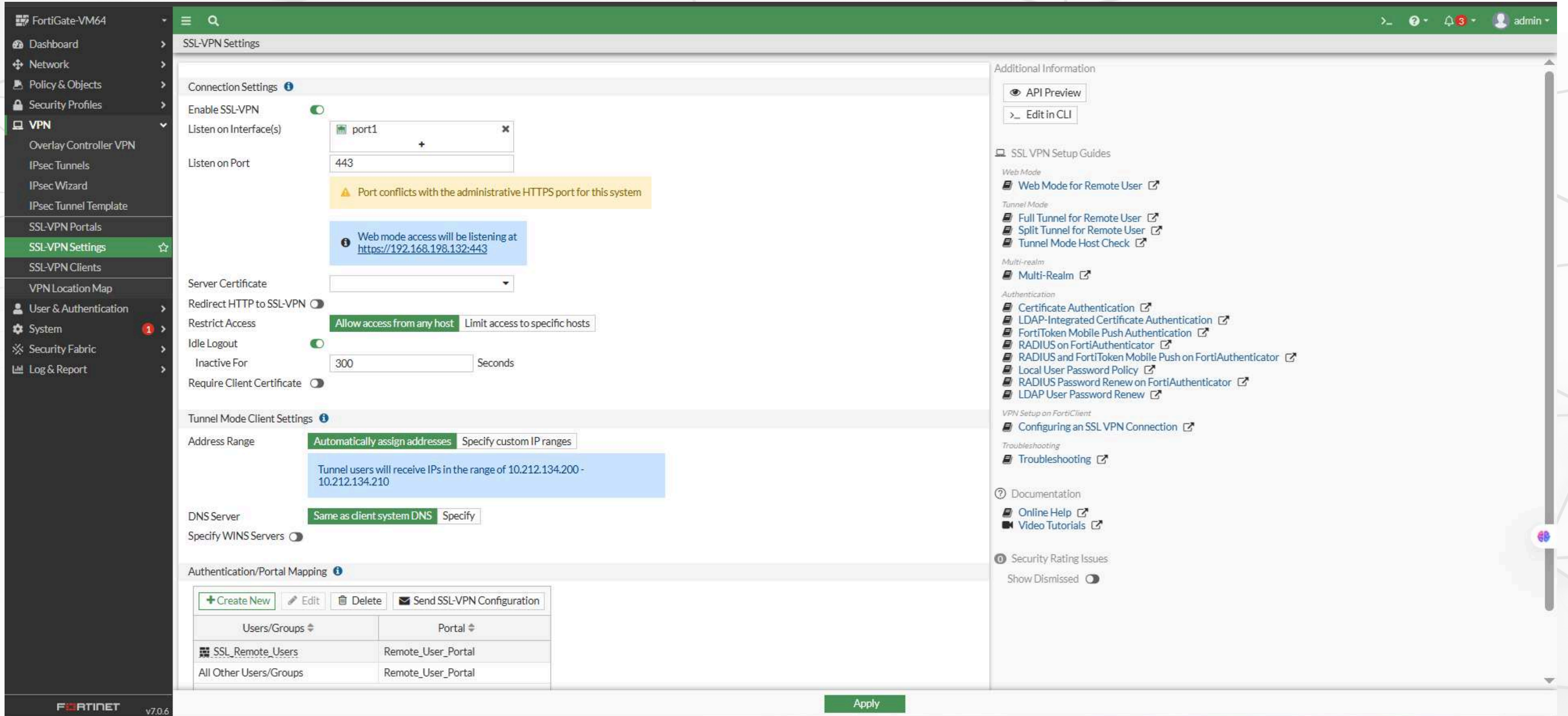


The screenshot shows the FortiGate VM64 web interface. The left sidebar menu is expanded to 'User & Authentication', with 'User Groups' selected. The main content area displays a table of user groups. The table has four columns: Group Name, Group Type, Members, and Ref. The 'SSL_Remote_Users' group is highlighted, showing it is of type 'Firewall' and has two members, including 'yasmin_cli'.

Group Name	Group Type	Members	Ref.
Guest-group	Firewall	guest	0
SSL_Remote_Users	Firewall	yasmin_cli	2
SSO_Guest_Users	Fortinet Single Sign-On (FSSO)		1

Step 4 – SSL VPN Settings:

- SSL-VPN was enabled on port1.



The screenshot displays the FortiGate-VM64 SSL-VPN Settings page. The left sidebar shows the navigation menu with 'VPN' expanded and 'SSL-VPN Settings' selected. The main content area is divided into 'Connection Settings' and 'Tunnel Mode Client Settings'.

Connection Settings:

- Enable SSL-VPN:** Enabled (toggle switch).
- Listen on Interface(s):** port1
- Listen on Port:** 443
- Warning:** Port conflicts with the administrative HTTPS port for this system.
- Information:** Web mode access will be listening at <https://192.168.198.132:443>
- Server Certificate:** (Dropdown menu)
- Redirect HTTP to SSL-VPN:** Disabled (toggle switch).
- Restrict Access:** Allow access from any host (selected) | Limit access to specific hosts
- Idle Logout:** Enabled (toggle switch).
- Inactive For:** 300 Seconds
- Require Client Certificate:** Disabled (toggle switch).

Tunnel Mode Client Settings:

- Address Range:** Automatically assign addresses (selected) | Specify custom IP ranges
- Information:** Tunnel users will receive IPs in the range of 10.212.134.200 - 10.212.134.210
- DNS Server:** Same as client system DNS (selected) | Specify
- Specify WINS Servers:** Disabled (toggle switch).

Authentication/Portal Mapping:

Users/Groups	Portal
SSL_Remote_Users	Remote_User_Portal
All Other Users/Groups	Remote_User_Portal

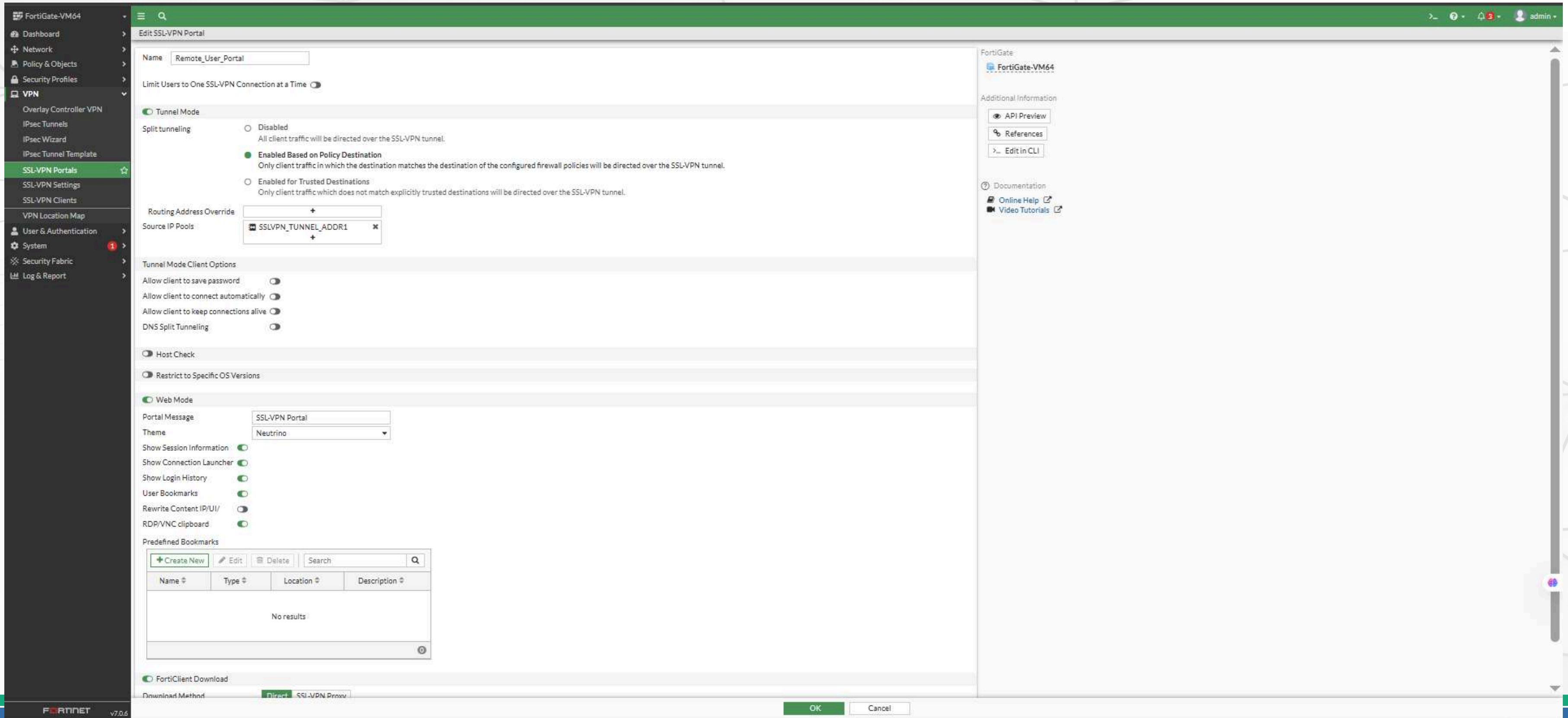
Additional Information:

- [API Preview](#)
- [Edit in CLI](#)
- SSL VPN Setup Guides:**
 - Web Mode:**
 - [Web Mode for Remote User](#)
 - Tunnel Mode:**
 - [Full Tunnel for Remote User](#)
 - [Split Tunnel for Remote User](#)
 - [Tunnel Mode Host Check](#)
 - Multi-realm:**
 - [Multi-Realm](#)
 - Authentication:**
 - [Certificate Authentication](#)
 - [LDAP-Integrated Certificate Authentication](#)
 - [FortiToken Mobile Push Authentication](#)
 - [RADIUS on FortiAuthenticator](#)
 - [RADIUS and FortiToken Mobile Push on FortiAuthenticator](#)
 - [Local User Password Policy](#)
 - [RADIUS Password Renew on FortiAuthenticator](#)
 - [LDAP User Password Renew](#)
 - VPN Setup on FortiClient:**
 - [Configuring an SSL VPN Connection](#)
 - Troubleshooting:**
 - [Troubleshooting](#)
 - Documentation:**
 - [Online Help](#)
 - [Video Tutorials](#)
 - Security Rating Issues:**
 - Show Dismissed

Fortinet v7.0.6 [Apply](#)

Step 5 – SSL-VPN Portal Setup:

- The Remote_User_Portal was configured.
- Split Tunneling was enabled to route internal traffic only.
- Routing override was set to the SSLVPN address pool.



The screenshot shows the FortiGate VM64 configuration interface for the SSL-VPN Portal. The left sidebar contains navigation options: Dashboard, Network, Policy & Objects, Security Profiles, VPN, Overlay Controller VPN, IPsec Tunnels, IPsec Wizard, IPsec Tunnel Template, SSL-VPN Portals (selected), SSL-VPN Settings, SSL-VPN Clients, VPN Location Map, User & Authentication, System, Security Fabric, and Log & Report.

The main configuration area is titled "Edit SSL-VPN Portal" and shows the following settings:

- Name:** Remote_User_Portal
- Limit Users to One SSL-VPN Connection at a Time:** ☐
- Tunnel Mode:** ☒ Tunnel Mode
 - Split tunneling:**
 - ☐ Disabled: All client traffic will be directed over the SSL-VPN tunnel.
 - ☒ Enabled Based on Policy Destination: Only client traffic in which the destination matches the destination of the configured firewall policies will be directed over the SSL-VPN tunnel.
 - ☐ Enabled for Trusted Destinations: Only client traffic which does not match explicitly trusted destinations will be directed over the SSL-VPN tunnel.
 - Routing Address Override:**
 - Source IP Pools:**
- Tunnel Mode Client Options:**
 - Allow client to save password:** ☐
 - Allow client to connect automatically:** ☐
 - Allow client to keep connections alive:** ☐
 - DNS Split Tunneling:** ☐
- Host Check:** ☐
- Restrict to Specific OS Versions:** ☐
- Web Mode:** ☒ Web Mode
 - Portal Message:** SSL-VPN Portal
 - Theme:** Neutrino
 - Show Session Information:** ☒
 - Show Connection Launcher:** ☒
 - Show Login History:** ☒
 - User Bookmarks:** ☒
 - Rewrite Content IP/UI/:** ☐
 - RDP/VNC clipboard:** ☒
- Predefined Bookmarks:**

Name	Type	Location	Description
No results			
- FortiClient Download:** ☒ FortiClient Download
 - Download Method:** Direct SSL-VPN Proxy

The right sidebar contains additional information links: API Preview, References, Edit in CLI, Documentation, Online Help, and Video Tutorials.

At the bottom of the interface, there are "OK" and "Cancel" buttons.

Step 6 – Firewall Policies:

- Policy created to allow LAN → SSLVPN internal communication.
- Policy created to allow SSLVPN → LAN user access to internal resources.

FortiGate-VM64

Dashboard Network Policy & Objects Firewall Policy IPv4 DoS Policy Addresses Internet Service Database Services Schedules Virtual IPs IP Pools Protocol Options Traffic Shaping Security Profiles VPN User & Authentication System Security Fabric Log & Report

Create New Edit Delete Policy Lookup Search

Export Interface Pair View By Sequence

Name	Source	Destination	Schedule	Service	Action	NAT	Security Profiles	Log	Bytes
port2 → SSL-VPN tunnel interface (ssl.root) 1									
LAN_to_SSLVPN	LAN_Subnet	SSLVPN_TUNNEL_ADDR1	always	ALL	✓ ACCEPT	✗ Disabled	SSL no-inspection	✓ All	0 B
SSL-VPN tunnel interface (ssl.root) → port2 1									
SSLVPN_to_LAN	SSL_Remote_Users SSLVPN_TUNNEL_ADDR1	LAN_Subnet	always	ALL	✓ ACCEPT	✗ Disabled	SSL no-inspection	UTM	0 B
Implicit 1									

Security Rating Issues

Updated: 10:06:35

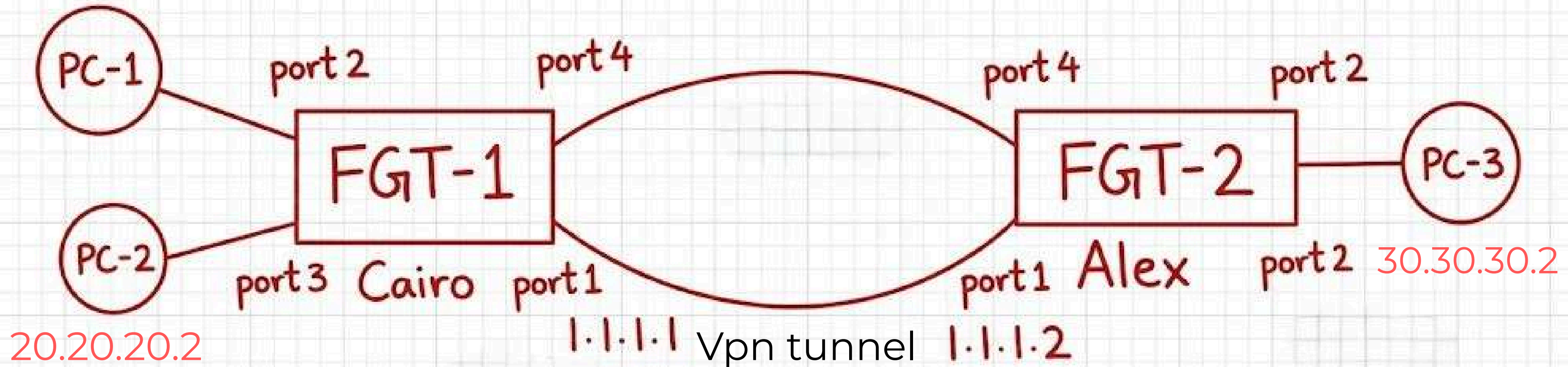
Conclusion

All SSL-VPN configurations required for Week 1 have been successfully completed, including interface setup, user and group creation, SSL-VPN settings, portal configuration, and firewall policies. Connection testing was not performed at this stage due to the expected limitations of the FortiGate VM edition, but the configuration is fully correct and ready for testing once a licensed environment is available.

Week2: IPsec Site-to-Site VPN Tunnel Implementation

Topology Overview

10.10.10.2



Goal Traffic:

Secure communication primarily between the 10.10.10.0/24 and 30.30.30.0/24 networks.

Configuration

Step 1 – Interface Configuration:

- Port1 was configured as the management interface on both firewalls.
- IP addresses were assigned to all required interfaces to provide both internal and external connectivity.
- Each site was configured with its correct internal subnets and WAN interfaces.

Cairo

Dashboard

Network

Interfaces

DNS

Packet Capture

SD-WAN

Static Routes

Policy Routes

RIP

OSPF

BGP

Routing Objects

Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

FortiGate VM64

1 3 5 7 9 11 13 15 17 19 21 23

2 4 6 8 10 12 14 16 18 20 22 24

Create New

Edit

Delete

Integrate Interface

Search

Group By Type

Name	Type	Members	IP/Netmask	Administrative Access	DHCP Clients	DHCP Ranges
802.3ad Aggregate						
fortilink	802.3ad Aggregate		Dedicated to FortiSwitch	PING Security Fabric Connection		10.255.1.2-10.255.1.254
Physical Interface						
LAN 1 (port2)	Physical Interface		10.10.10.1/255.255.255.0	PING	1	10.10.10.2-10.10.10.254
LAN 2 (port3)	Physical Interface		20.20.20.1/255.255.255.0	PING	1	20.20.20.2-20.20.20.254
port1	Physical Interface		192.168.1.13/255.255.255.0	PING HTTPS SSH HTTP FMG-Access		
port5	Physical Interface		0.0.0.0/0.0.0.0			
port6	Physical Interface		0.0.0.0/0.0.0.0			
port7	Physical Interface		0.0.0.0/0.0.0.0			

Security Rating Issues

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Updated: 05:42:56

Alex

Dashboard

Network

Interfaces

DNS

Packet Capture

SD-WAN

Static Routes

Policy Routes

RIP

OSPF

BGP

Routing Objects

Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

FortiGate VM64

1 3 5 7 9 11 13 15 17 19 21 23

2 4 6 8 10 12 14 16 18 20 22 24

Create New Edit Delete Integrate Interface

Search

Group By Type

Name	Type	Members	IP/Netmask	Administrative Access	DHCP Clients	DHCP Ranges
802.3ad Aggregate						
fortilink	802.3ad Aggregate		Dedicated to FortiSwitch	PING Security Fabric Connection		10.255.1.2-10.255.1.254
Physical Interface						
LAN (port2)	Physical Interface		30.30.30.1/255.255.255.0	PING	1	30.30.30.2-30.30.30.254
port1	Physical Interface		192.168.1.11/255.255.255.0	PING HTTPS SSH HTTP FMG-Access		
port3	Physical Interface		0.0.0.0/0.0.0.0			
port5	Physical Interface		0.0.0.0/0.0.0.0			
port6	Physical Interface		0.0.0.0/0.0.0.0			
port7	Physical Interface		0.0.0.0/0.0.0.0			

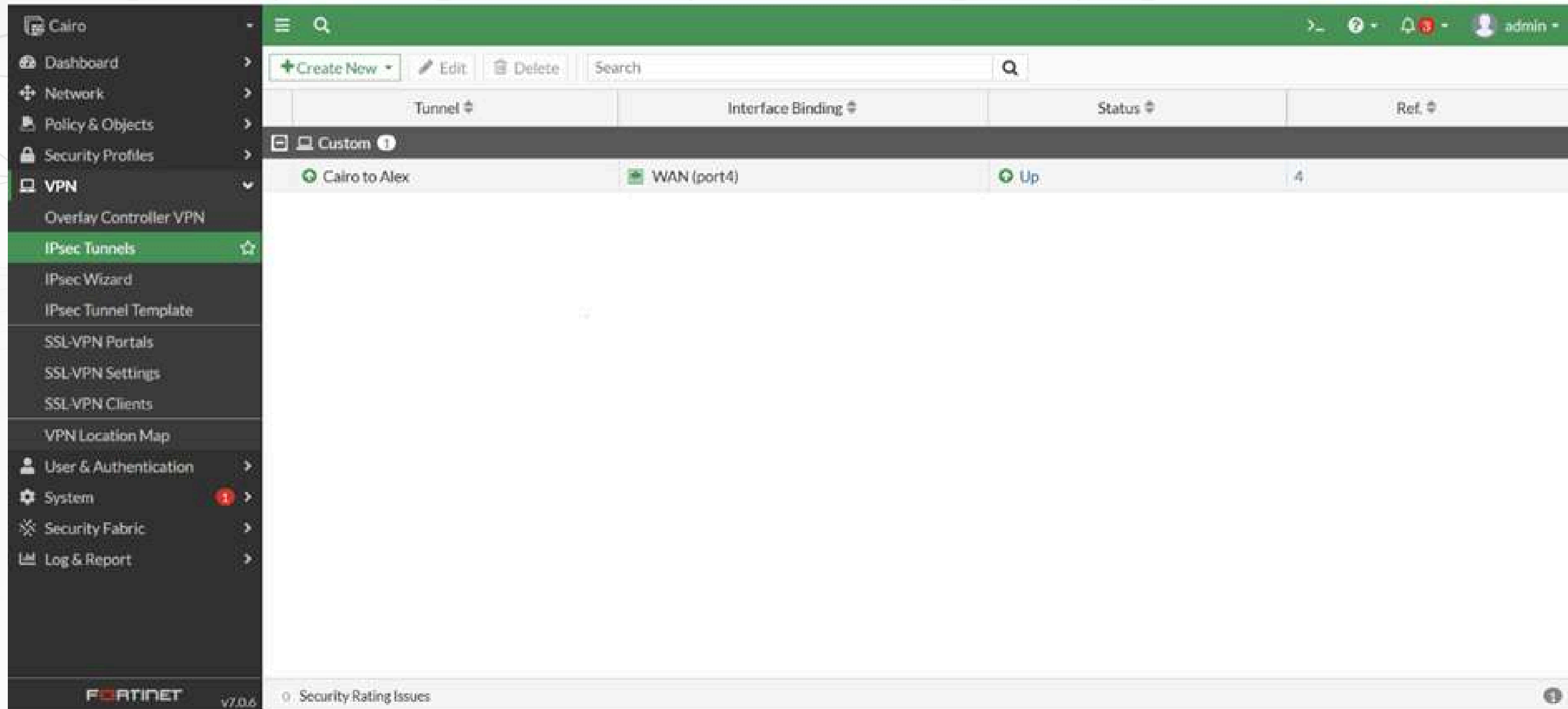
Security Rating Issues

0% Updated: 05:51:31

Configuration

Step 2 – VPN Tunnel Configuration

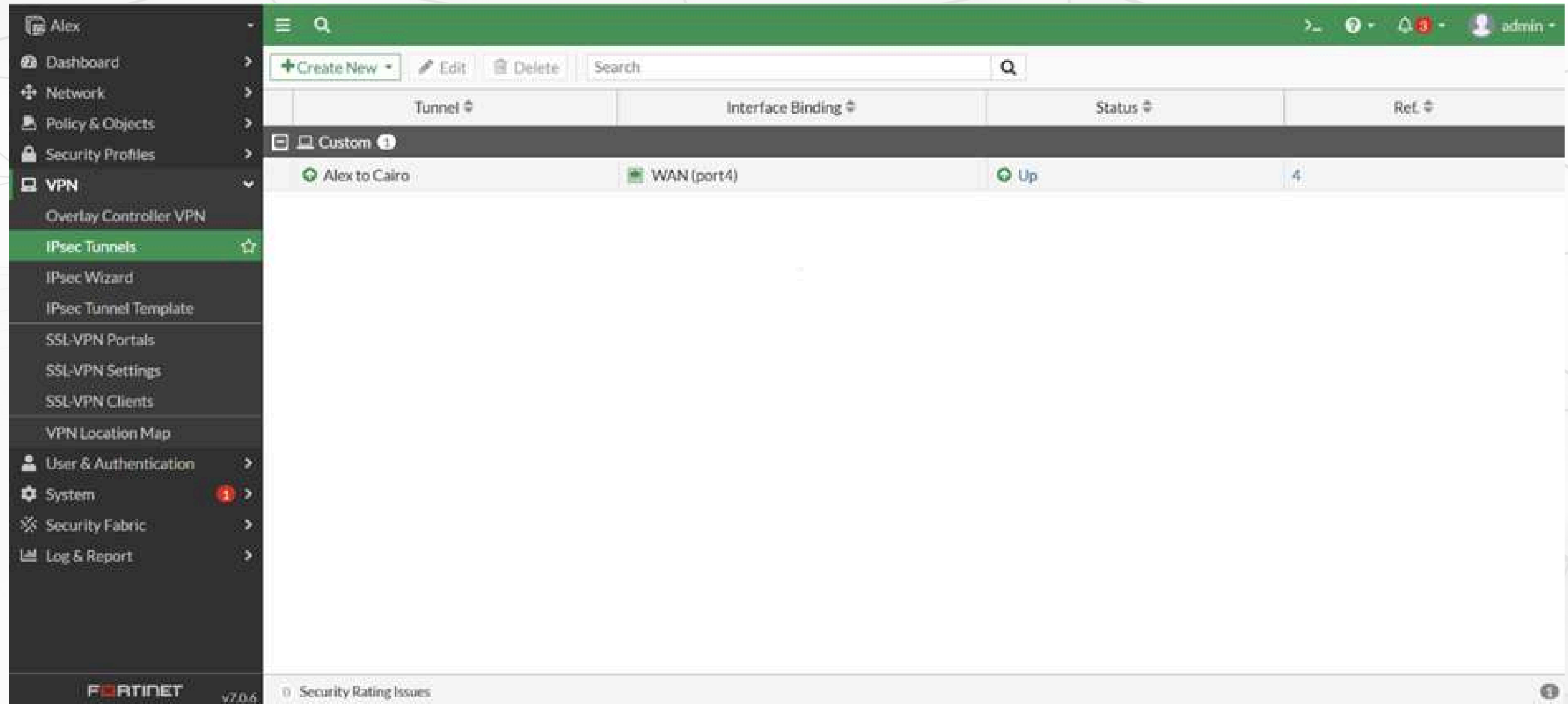
- An IPsec VPN tunnel was created on both firewalls.
- The traffic selectors were configured as follows:
 - Cairo: 10.10.10.0/24 → 30.30.30.0/24
 - Alex: 30.30.30.0/24 → 10.10.10.0/24
- Phase 1 and Phase 2 parameters were matched to ensure successful negotiation of the VPN tunnel.



The image shows the Fortinet FortiGate v7.0.6 web interface for configuring VPNs. The left sidebar contains a navigation menu with the following items: Cairo, Dashboard, Network, Policy & Objects, Security Profiles, VPN (expanded), Overlay Controller VPN, IPsec Tunnels (highlighted), IPsec Wizard, IPsec Tunnel Template, SSL-VPN Portals, SSL-VPN Settings, SSL-VPN Clients, VPN Location Map, User & Authentication, System (with a red notification badge), Security Fabric, and Log & Report. The main content area displays the IPsec Tunnels configuration page. At the top, there is a green header bar with a search icon and a user profile labeled 'admin'. Below this is a toolbar with 'Create New', 'Edit', 'Delete', and a search field. The main table lists the configured tunnels:

Tunnel	Interface Binding	Status	Ref.
Custom 1			
Cairo to Alex	WAN (port4)	Up	4

The bottom status bar shows '0 Security Rating Issues' and the FortiGate logo with version 'v7.0.6'.



The screenshot displays the Fortinet FortiGate web interface. On the left is a dark sidebar menu with the following items: Alex, Dashboard, Network, Policy & Objects, Security Profiles, VPN (expanded), Overlay Controller VPN, IPsec Tunnels (highlighted), IPsec Wizard, IPsec Tunnel Template, SSL-VPN Portals, SSL-VPN Settings, SSL-VPN Clients, VPN Location Map, User & Authentication, System (with a red notification badge), Security Fabric, and Log & Report. The main content area has a green header bar with a search icon and a user profile 'admin'. Below the header is a toolbar with '+ Create New', 'Edit', 'Delete', and a search field. The main table lists IPsec tunnels with columns: Tunnel, Interface Binding, Status, and Ref. One tunnel is listed: 'Alex to Cairo' with interface 'WAN (port4)' and status 'Up'. The bottom status bar shows '0 Security Rating Issues' and the FortiGate version 'v7.0.6'.

Tunnel	Interface Binding	Status	Ref.
Alex to Cairo	WAN (port4)	Up	4

Configuration

Step 3 – Static Route Configuration

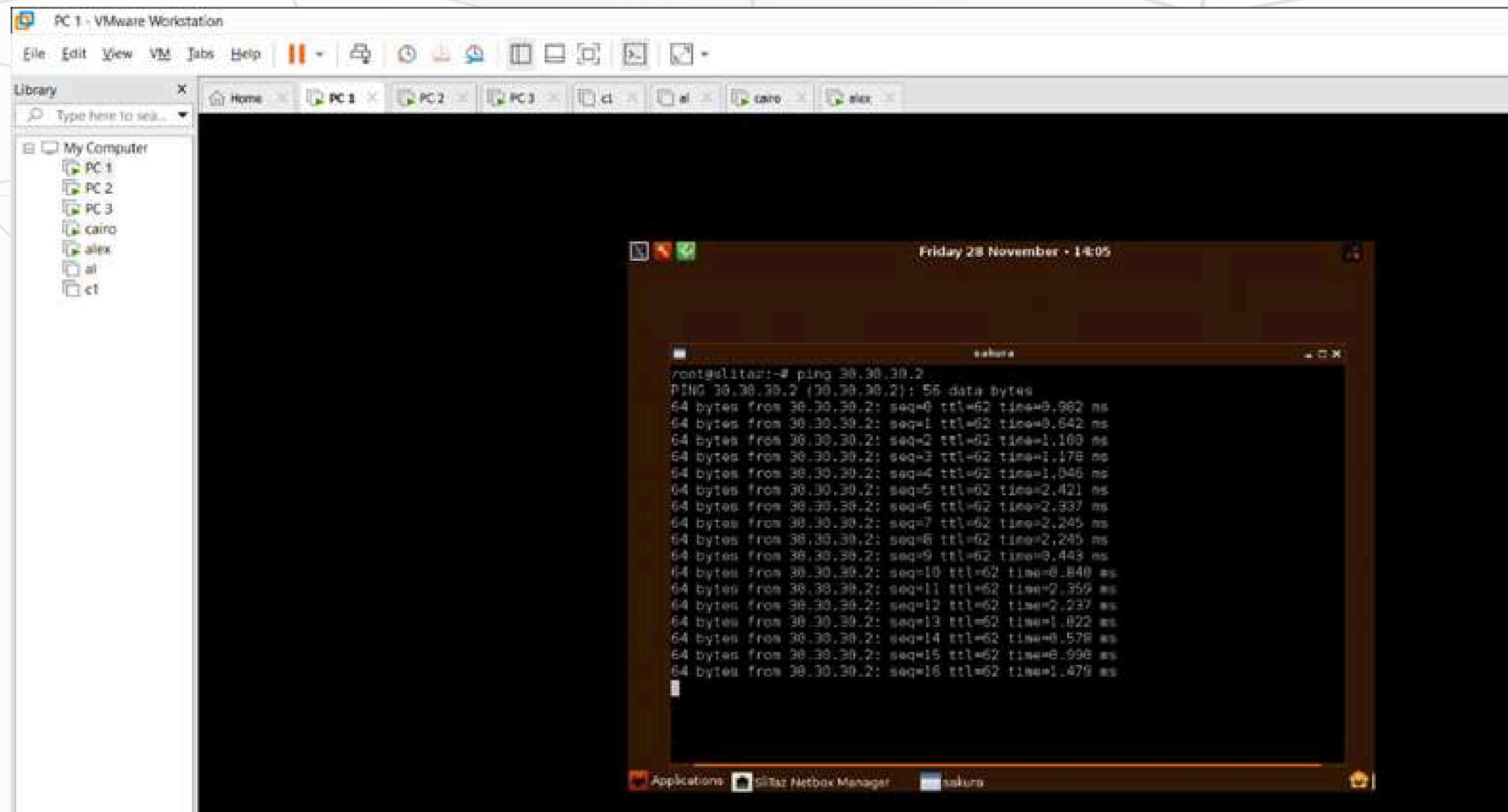
- Static routes were added on each firewall so that traffic destined for the remote network is forwarded into the VPN tunnel.
- This ensures correct routing between the Cairo and Alex networks.

Step 4 – Firewall Policy Configuration

- Security policies were created to allow traffic between the internal networks and the VPN tunnel interface.
- Only traffic that matches the defined traffic selector is allowed through the tunnel.

Connectivity Test Results

Ping Test: PC-1 to PC-3

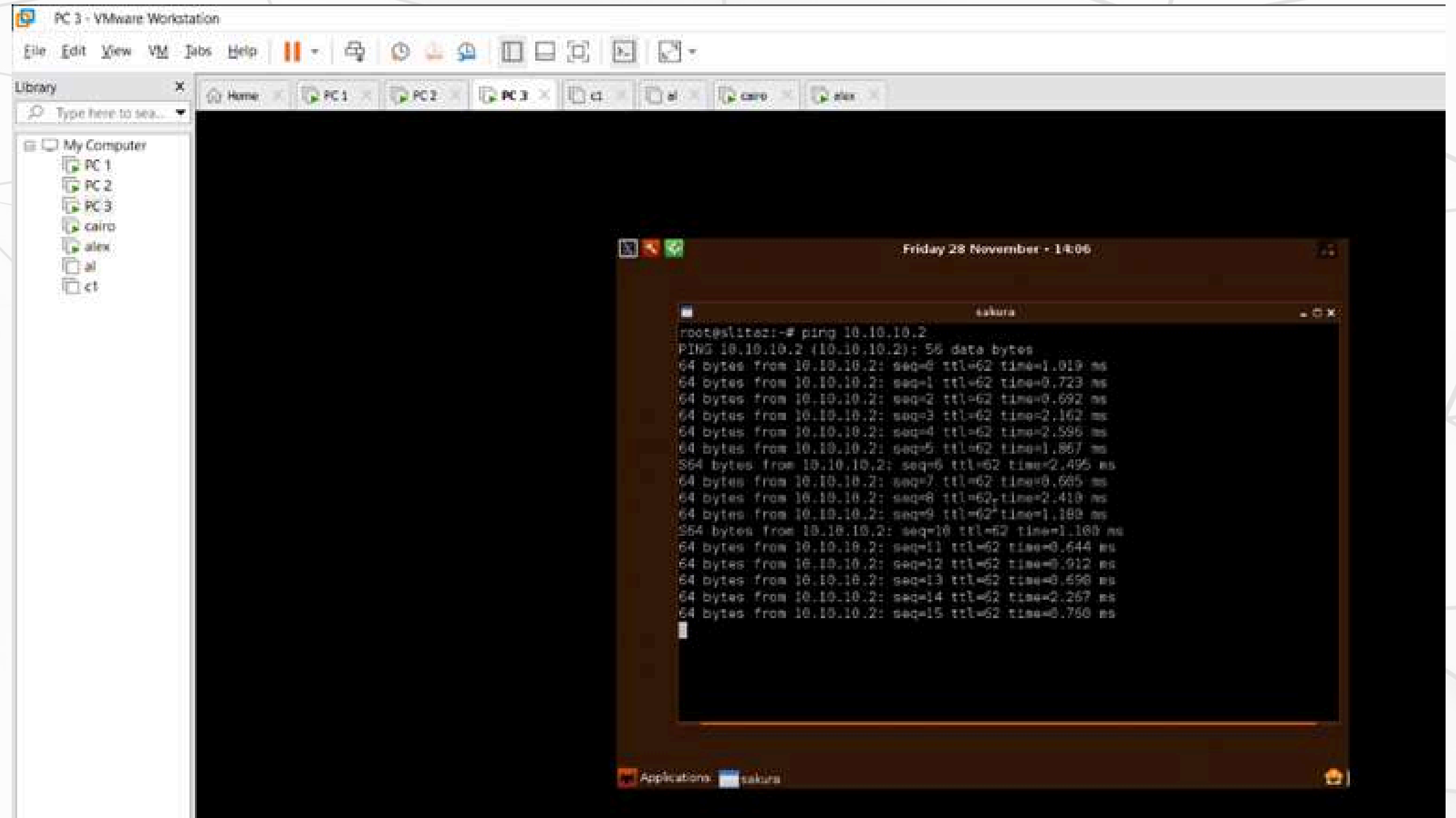


```
PC 1 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
PC 1
PC 2
PC 3
cairo
alex
al
cl

Friday 28 November - 14:05
root@slitaz:~# ping 30.30.30.2
PING 30.30.30.2 (30.30.30.2): 56 data bytes
64 bytes from 30.30.30.2: seq=0 ttl=62 time=0.962 ms
64 bytes from 30.30.30.2: seq=1 ttl=62 time=0.642 ms
64 bytes from 30.30.30.2: seq=2 ttl=62 time=1.100 ms
64 bytes from 30.30.30.2: seq=3 ttl=62 time=1.178 ms
64 bytes from 30.30.30.2: seq=4 ttl=62 time=1.046 ms
64 bytes from 30.30.30.2: seq=5 ttl=62 time=2.421 ms
64 bytes from 30.30.30.2: seq=6 ttl=62 time=2.337 ms
64 bytes from 30.30.30.2: seq=7 ttl=62 time=2.345 ms
64 bytes from 30.30.30.2: seq=8 ttl=62 time=2.245 ms
64 bytes from 30.30.30.2: seq=9 ttl=62 time=0.443 ms
64 bytes from 30.30.30.2: seq=10 ttl=62 time=0.840 ms
64 bytes from 30.30.30.2: seq=11 ttl=62 time=2.359 ms
64 bytes from 30.30.30.2: seq=12 ttl=62 time=2.237 ms
64 bytes from 30.30.30.2: seq=13 ttl=62 time=1.022 ms
64 bytes from 30.30.30.2: seq=14 ttl=62 time=0.576 ms
64 bytes from 30.30.30.2: seq=15 ttl=62 time=0.996 ms
64 bytes from 30.30.30.2: seq=16 ttl=62 time=1.479 ms
```

Connectivity Test Results

Ping Test: PC-3 to PC-1



The screenshot shows a VMware Workstation window titled "PC 3 - VMware Workstation". The interface includes a menu bar (File, Edit, View, VM, Tabs, Help), a toolbar, and a Library pane on the left. The Library pane shows a tree structure under "My Computer" with items: PC 1, PC 2, PC 3, cairo, alex, al, and ct. The main window displays a terminal window titled "sakura" with the following output:

```
Friday 28 November - 14:06
root@slitez:~# ping 10.10.10.2
PING 10.10.10.2 (10.10.10.2): 56 data bytes
64 bytes from 10.10.10.2: seq=0 ttl=62 time=1.019 ms
64 bytes from 10.10.10.2: seq=1 ttl=62 time=0.723 ms
64 bytes from 10.10.10.2: seq=2 ttl=62 time=0.692 ms
64 bytes from 10.10.10.2: seq=3 ttl=62 time=2.162 ms
64 bytes from 10.10.10.2: seq=4 ttl=62 time=2.596 ms
64 bytes from 10.10.10.2: seq=5 ttl=62 time=1.867 ms
364 bytes from 10.10.10.2: seq=6 ttl=62 time=2.495 ms
64 bytes from 10.10.10.2: seq=7 ttl=62 time=0.685 ms
64 bytes from 10.10.10.2: seq=8 ttl=62 time=2.410 ms
64 bytes from 10.10.10.2: seq=9 ttl=62 time=1.180 ms
254 bytes from 10.10.10.2: seq=10 ttl=62 time=1.180 ms
64 bytes from 10.10.10.2: seq=11 ttl=62 time=0.644 ms
64 bytes from 10.10.10.2: seq=12 ttl=62 time=0.912 ms
64 bytes from 10.10.10.2: seq=13 ttl=62 time=0.498 ms
64 bytes from 10.10.10.2: seq=14 ttl=62 time=2.267 ms
64 bytes from 10.10.10.2: seq=15 ttl=62 time=0.790 ms
```

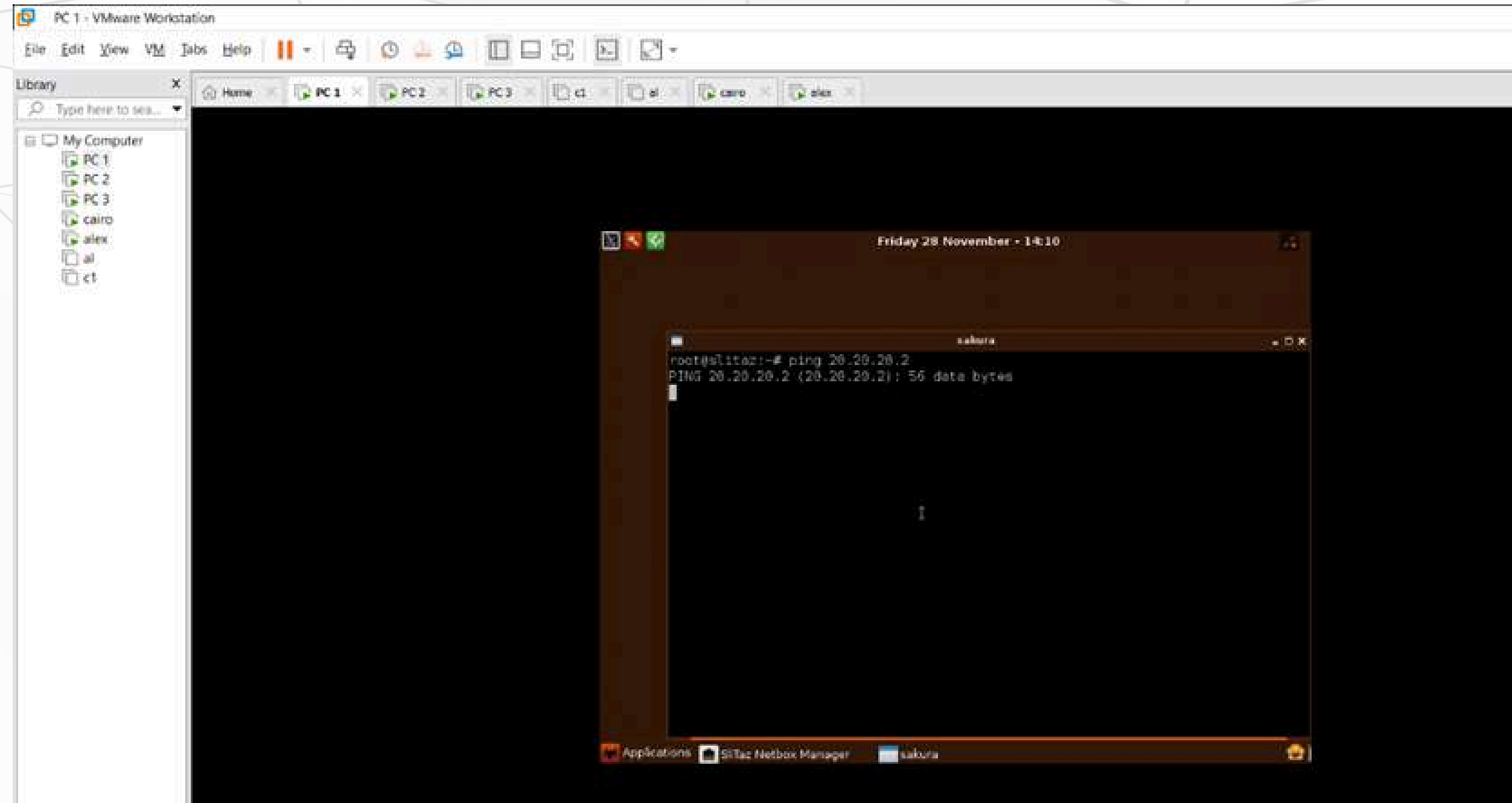
Connectivity Test Results

Comment:

- The ping was successful, confirming that the IPsec VPN tunnel is working correctly.
- Connectivity is allowed because the traffic selector includes 10.10.10.0/24 ↔ 30.30.30.0/24.

Connectivity Test Results

Ping Test: PC-1 to PC-2



Connectivity Test Results

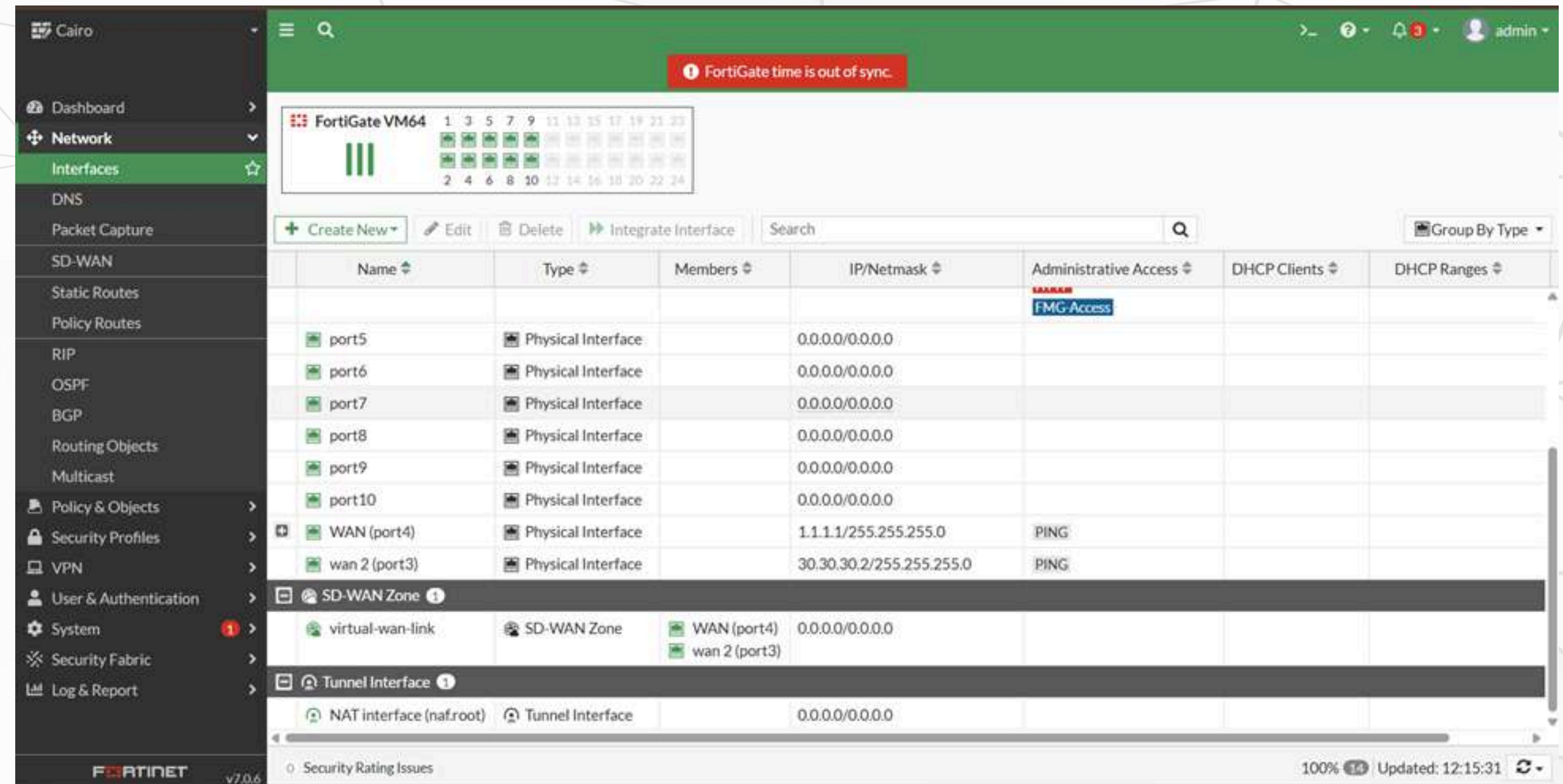
Comment:

- The ping failed, which is the expected result.
- No traffic selector is configured between the 10 network and the 20 network, so the firewall drops this traffic.

Week 3: FortiGate VPN with SD-WAN

Configuration steps of SD-WAN on FortiGate

Figure 1:configure the ports to a wan



FortiGate VM64

FortiGate time is out of sync.

Network

Interfaces

DNS

Packet Capture

SD-WAN

Static Routes

Policy Routes

RIP

OSPF

BGP

Routing Objects

Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

FortiGate VM64

1 3 5 7 9 11 13 15 17 19 21 23

2 4 6 8 10 12 14 16 18 20 22 24

Create New Edit Delete Integrate Interface Search

Name	Type	Members	IP/Netmask	Administrative Access	DHCP Clients	DHCP Ranges
port5	Physical Interface		0.0.0.0/0.0.0.0			
port6	Physical Interface		0.0.0.0/0.0.0.0			
port7	Physical Interface		0.0.0.0/0.0.0.0			
port8	Physical Interface		0.0.0.0/0.0.0.0			
port9	Physical Interface		0.0.0.0/0.0.0.0			
port10	Physical Interface		0.0.0.0/0.0.0.0			
WAN (port4)	Physical Interface		1.1.1.1/255.255.255.0	PING		
wan 2 (port3)	Physical Interface		30.30.30.2/255.255.255.0	PING		
SD-WAN Zone						
virtual-wan-link	SD-WAN Zone	WAN (port4) wan 2 (port3)	0.0.0.0/0.0.0.0			
Tunnel Interface						
NAT interface (naf.root)	Tunnel Interface		0.0.0.0/0.0.0.0			

Security Rating Issues

100% Updated: 12:15:31

Figure 2 : create new SD-WAN member

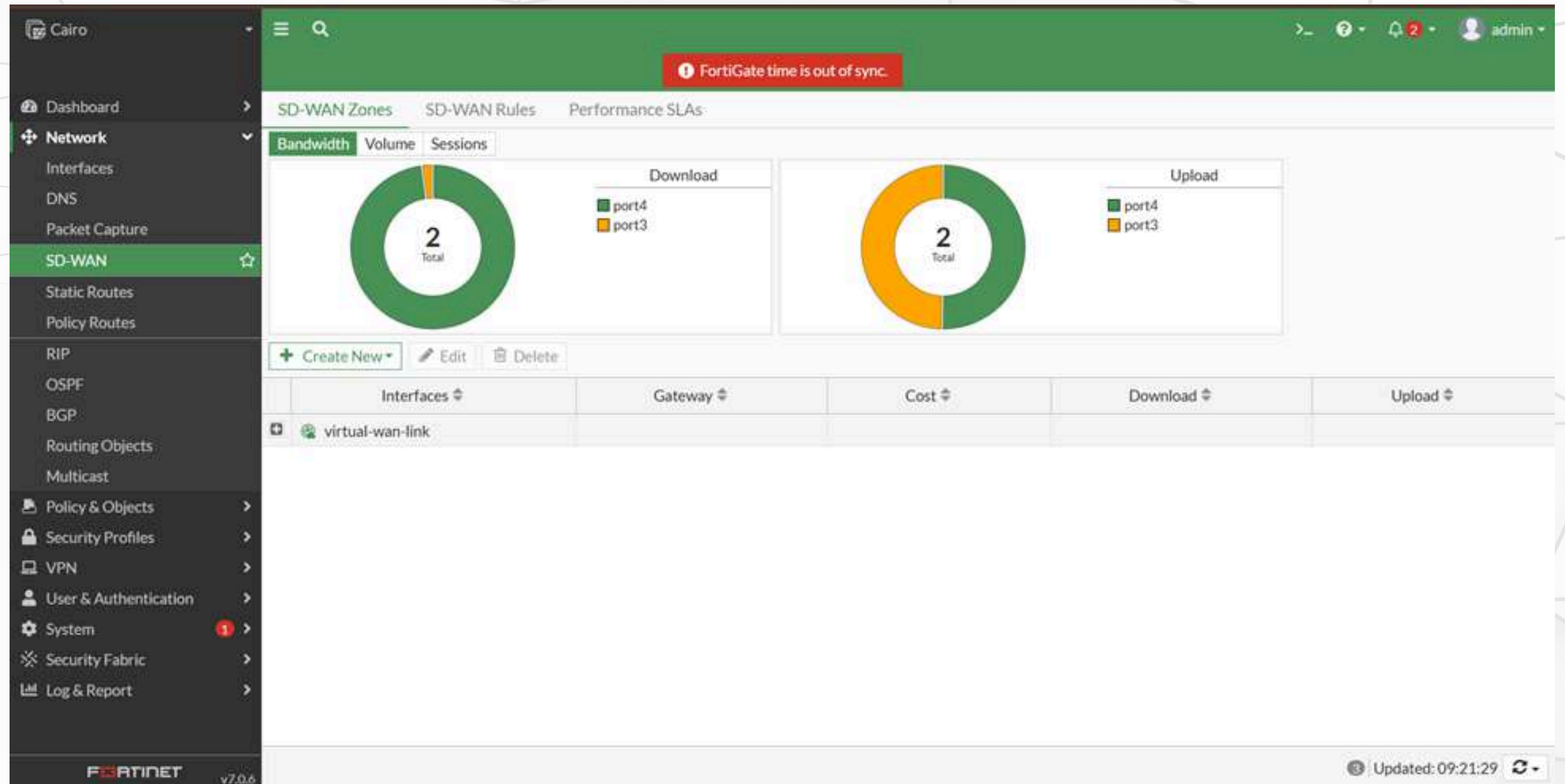
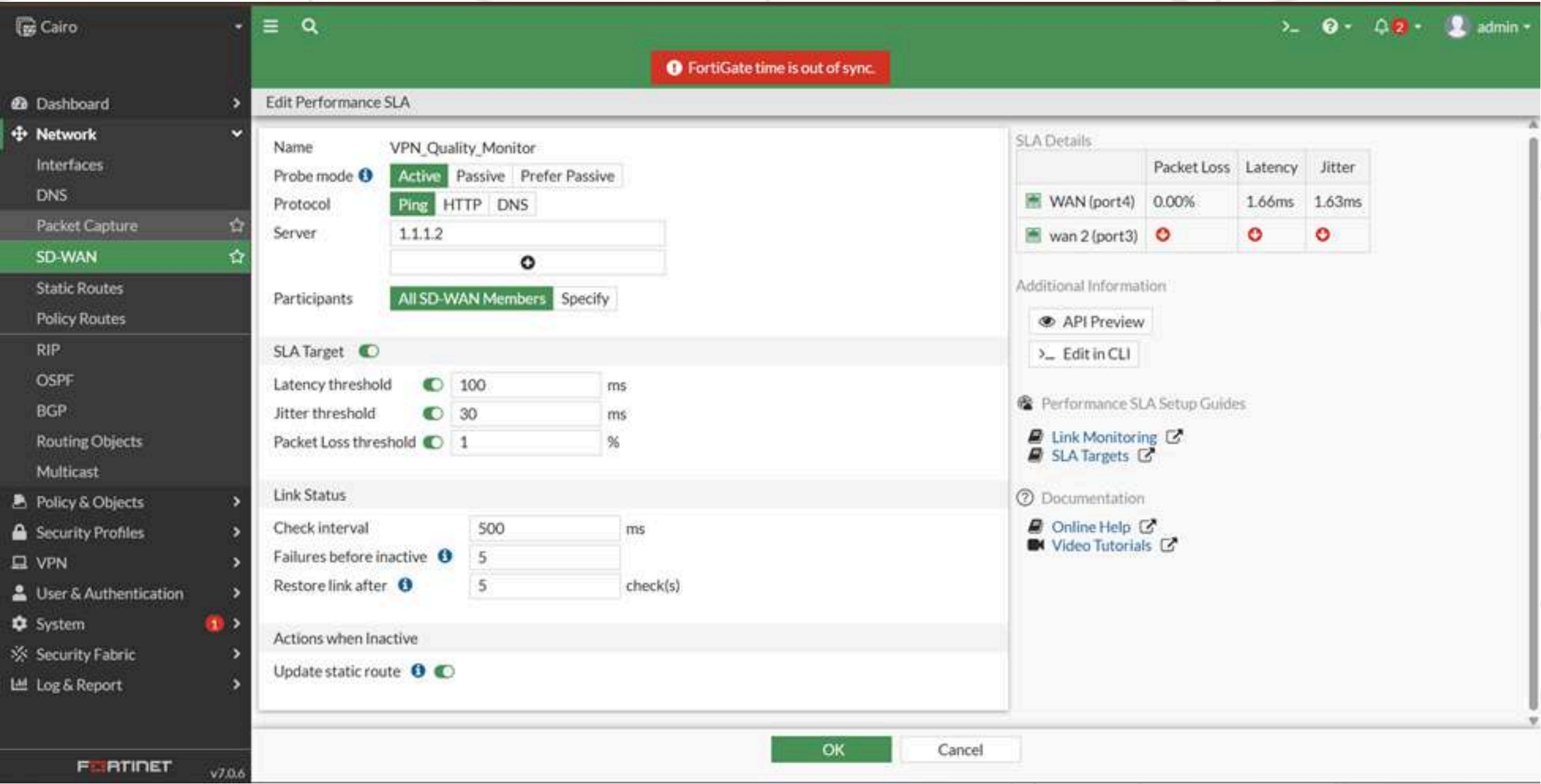


Figure 3: Define Link Performance Metrics (Performance SLA)

Purpose: This is the core intelligence. You define the acceptable quality targets (latency, jitter, loss) for the link carrying the VPN traffic. The FortiGate constantly checks both wan1 and wan2 against these targets.



The screenshot shows the FortiGate SD-WAN configuration interface for 'Edit Performance SLA'. The left sidebar shows the navigation menu with 'SD-WAN' selected. The main panel displays the configuration for 'VPN_Quality_Monitor'.

Configuration Details:

- Name:** VPN_Quality_Monitor
- Probe mode:** Active (selected), Passive, Prefer Passive
- Protocol:** Ping (selected), HTTP, DNS
- Server:** 1.1.1.2
- Participants:** All SD-WAN Members (selected), Specify
- SLA Target:** Enabled (toggle)
- SLA Thresholds:**
 - Latency threshold: 100 ms
 - Jitter threshold: 30 ms
 - Packet Loss threshold: 1 %
- Link Status:**
 - Check interval: 500 ms
 - Failures before inactive: 5
 - Restore link after: 5 check(s)
- Actions when Inactive:**
 - Update static route: Enabled (toggle)

SLA Details Table:

	Packet Loss	Latency	Jitter
WAN (port4)	0.00%	1.66ms	1.63ms
wan 2 (port3)	⬇	⬇	⬇

Additional Information:

- API Preview
- Edit in CLI
- Performance SLA Setup Guides
 - Link Monitoring
 - SLA Targets
- Documentation
 - Online Help
 - Video Tutorials

Buttons: OK, Cancel

Figure 4 : Create the SD-WAN Rule (Traffic Steering Logic)

Purpose: This rule specifically directs the VPN traffic to use the best performing link based on the real-time data collected

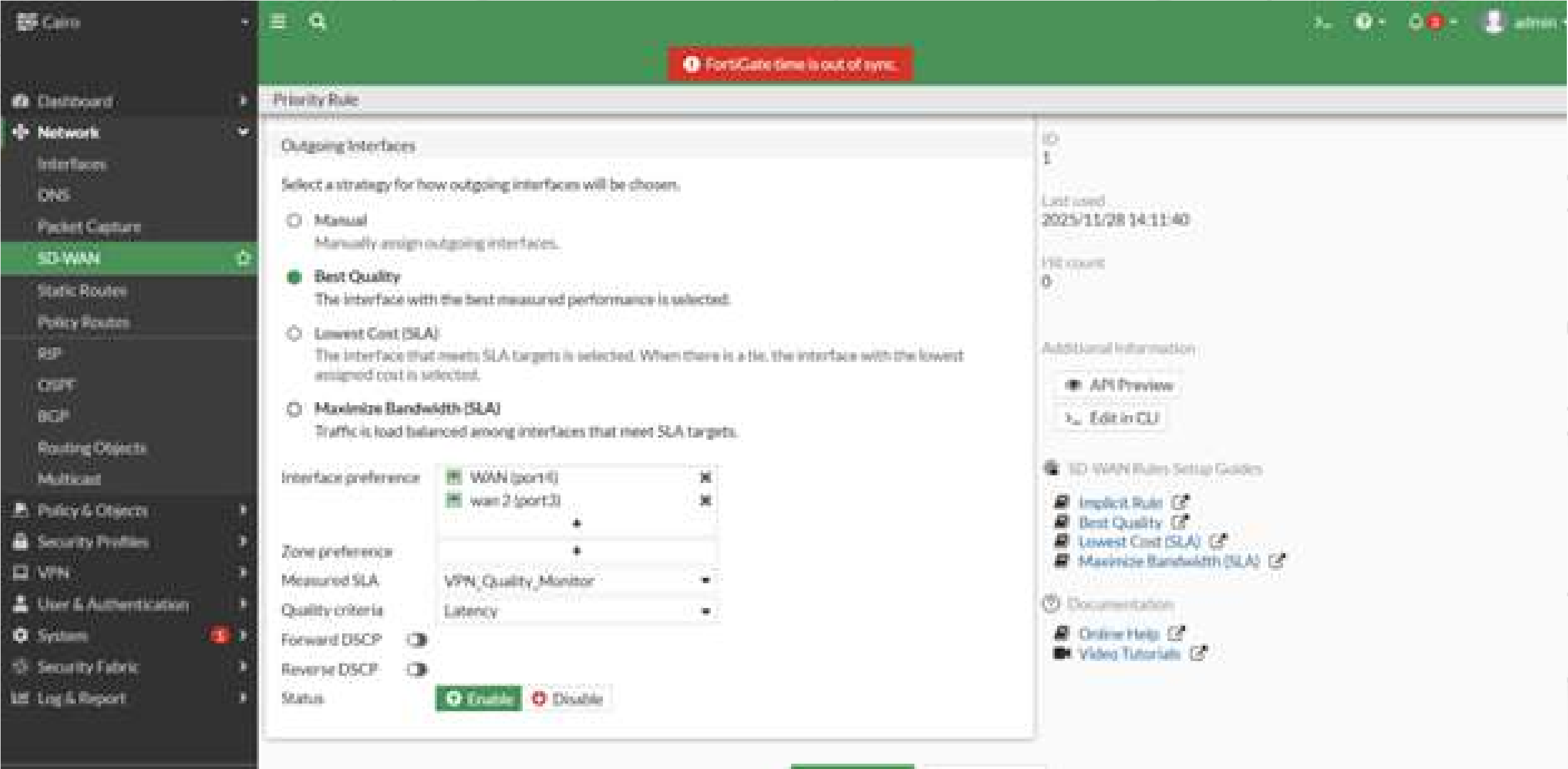
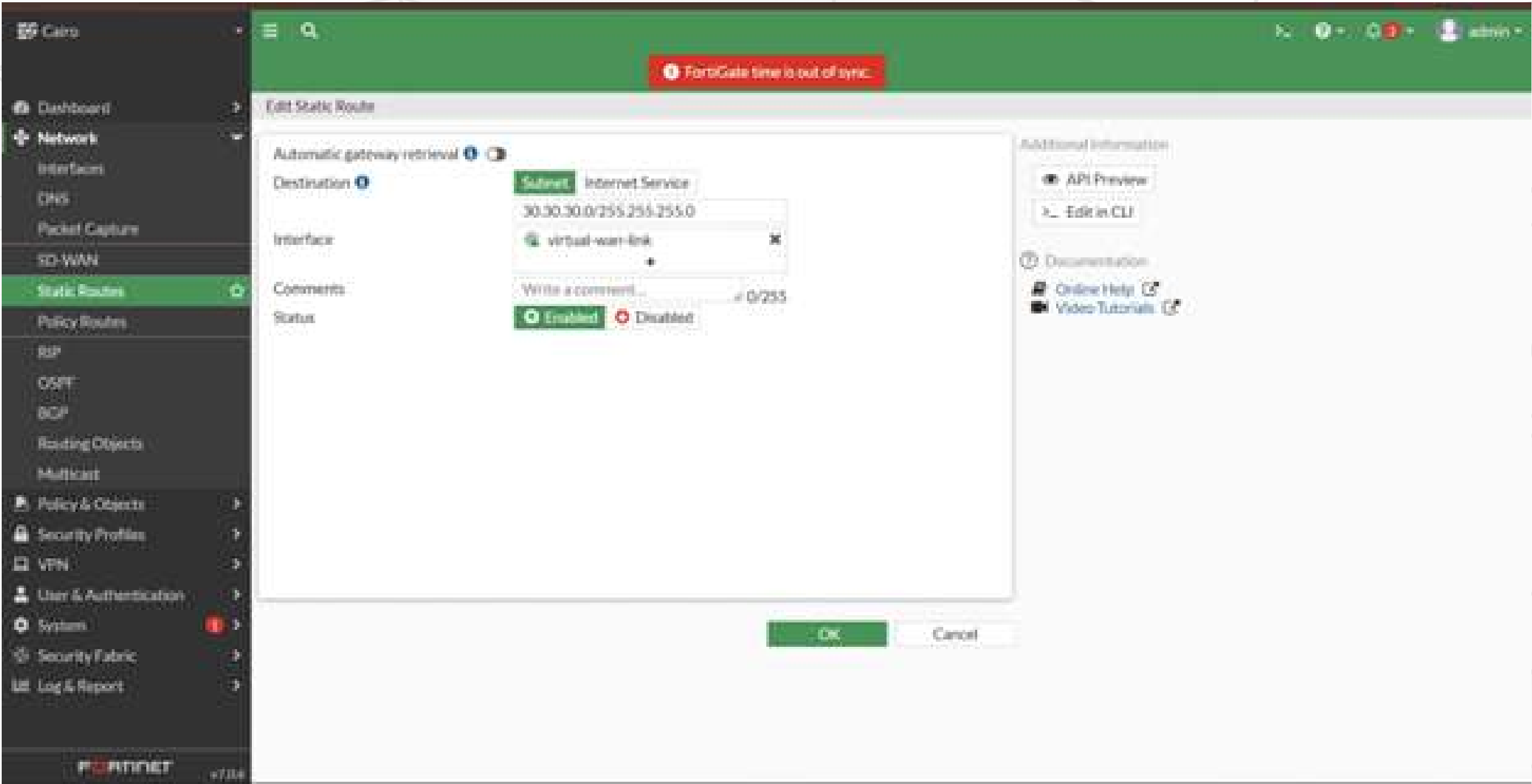


Figure 5 : Update the Static Route

Purpose: To make the SD-WAN rule effective, you must redirect the traffic bound for the remote VPN network away from a single physical interface and hand it over to the SD-WAN engine.



Conclusion

- Successfully implemented three FortiGate VPN solutions: **SSL, IPsec, and SD-WAN**.
- **SSL VPN** → Delivered secure remote access with user authentication and controlled permissions.
- **IPsec VPN** → Enabled encrypted site-to-site communication and verified tunnel stability through testing.
- **SD-WAN integration** → Improved VPN reliability by monitoring link metrics and automatically selecting optimal paths.
- Demonstrated real **configuration, troubleshooting, and validation**, not just theoretical concepts.



Any Questions ?

**THANK
YOU!**